Methodology

Methodology I

Numbered-table references point to the canonical manuscript; they are not standalone local deck numbers.

Request and canonical artifact

Request and canonical artifact I

DuckRabbit starts from a typed request

$$q = (i, \theta, s, e) \tag{1}$$

where i is an illusion identifier, θ is an immutable parameter object, s is a deterministic seed, and e is an optional encoding request. A registered generator maps the request to a canonical artifact:

$$A = G_i(\theta; s), \qquad A \in \{I, X, V, AV\} \tag{2}$$

The four artifact types are an image frame I, audio buffer X, video sequence V, and audiovisual timeline AV. The generator core is independent of Pillow, WAV containers, GIF, MP4, and ffmpeg. Each artifact retains its shape, dtype, units, clock, channels, and timing offsets.

Typed parameter domains

Typed parameter domains I

Value objects validate finite ranges at construction time. Image parameters include dimensions, color mode, luminance bounds, grayscale levels, and quantization levels. Audio parameters include frequency, phase, amplitude, duration, envelope, sample rate, channel count, and PCM depth. Video parameters include dimensions, rational frame rate, frame count, and temporal offsets. Audiovisual parameters compose audio and video clocks with declared synchronization and spatial discrepancies.

Typed parameter domains I

The canonical sample count and frame timestamps are explicit. For audio duration T and sample rate f_s, N is rounded once at construction. For a video with frame rate f_r, frame index k is timestamped on the presentation clock. The audio-minus-video sign convention is explicit:

$$N = \mathrm{round}(f_s T), \qquad t_k = k / f_r, \qquad \Delta_{AV} = t_{\mathrm{audio}} - t_{\mathrm{video}} \tag{3}$$

The package rejects non-finite values, invalid shapes, out-of-range samples, aliasing frequencies, incompatible channel layouts, impossible frame rates, and contradictory encoding requests.

Canonical serialization and provenance

Canonical serialization and provenance I

Canonical arrays are contiguous little-endian float32 buffers with explicit metadata. Serialization includes the array values, shape, clock, and units; metadata are part of the identity rather than an informal annotation:

$$c(A) = \mathrm{Serialize}_{\mathrm{LE},\mathrm{float32}}(A, \mathrm{shape}, \mathrm{clock}, \mathrm{units}), \qquad h_c(A) = H(c(A)) \tag{4}$$

The canonical digest is independent of PNG, WAV, GIF, MP4, and NPZ delivery choices. The v2 manifest records the parameter schema, serialized parameters, taxonomy, evidence boundary, canonical facts, objective metrics, encoding profile, backend identity, decoded inspection, and verification status. A v1 reader remains available and marks upconverted records unverified until the encoded file is re-inspected.

Canonical serialization and provenance I

This separation follows a reproducibility principle from computational research: a result is not made reproducible merely by publishing a final file; the executable inputs, transformation steps, versioned software, and provenance must remain identifiable (Sandve et al. 2013; Wilson et al. 2014). The manifest therefore records the construction identity and delivery identity separately. A downstream user can cite the software version and regenerate the canonical object, or cite a particular encoded artifact when the delivery file itself is the relevant research object (Smith et al. 2016).

Canonical serialization and provenance I

The distinction also follows the older psychophysical problem of relating a controlled physical increment to a measured sensation. Fechner's 1860 treatise made measurement, stimulus control, and response comparison explicit parts of the scientific object (Fechner 1860). DuckRabbit adopts the stimulus-control and provenance aspects of that tradition, but it does not pretend that a canonical digest or media metric is a sensation measurement: observer data still require a task, presentation conditions, and a registered analysis.

Encoding and verification

Encoding and verification I

For an encoding request e, the delivered file F and decoded inspection I are:

$$F = E_e(A), \qquad I = D(F), \qquad V(F, M) \in \{\text{verified}, \text{failed}, \text{unverified}\} \tag{5}$$

The manifest M binds A, F, and I through hashes, typed summaries, and format-specific tolerances. PNG, GIF, WAV, and NPZ use local deterministic adapters. MP4 and muxed audiovisual output use an optional ffmpeg backend. Missing backends are reported as capability errors. Writes are atomic, parent directories are created, overwrite behavior is explicit, and a conflicting format argument is rejected.

Encoding and verification I

Canonical NPZ arrays must retain little-endian float32 storage; the reader and inspector reject float64, big-endian or integer replacements instead of silently coercing their values into a canonical-looking artifact.

For NPZ, verification replays declared metrics only under the supported computation, implementation-source and NumPy identities. Missing optional metrics receive no metric-verification claim; unsupported historical identities remain `unverified`, not relabeled current or falsely classified as corruption. A metric-specific successful check is required to infer measured-value agreement. Other encodings retain their disclosed quantization/decoded-fact boundary.

Objective metric definitions

Objective metric definitions I

`MetricRecord` and `MetricSuite` carry a metric name, value, unit, computation version, source artifact, tolerance, and claim level. Luminance statistics, quantization levels, RMS and peak amplitude, spectral summaries, frame deltas, stream durations, and synchronization offsets are physical or decoded-media facts. They are not observer responses and cannot by themselves establish an illusion effect. For canonical image samples $y_1, \ldots, y_n$, audio samples $x_1, \ldots, x_n$, and adjacent video frames Y_k, Y_{k+1}, the principal metrics are:

Objective metric definitions I

For images, y is the grayscale code value or the Rec. 709-weighted RGB channel-value proxy; no RGB transfer decoding or display calibration is assumed. The name `mean_luminance` is an API label, not a photometric measurement. For temporal differences, Y_k denotes the complete raw channel array, not the image luminance proxy. RGB audiovisual plots reuse the image proxy. Adjacent-frame subtraction and its float32 mean use C-contiguous scratch space, matching the canonical layout. This removes layout-dependent summary drift across exact NPZ round trips; Fortran/mixed-layout input rounding can therefore differ from older versions, without changing pixels or canonical byte identity.

Objective metric definitions I

$$\mu_Y = \frac{1}{n}\sum_{j=1}^{n} y_j, \qquad \sigma_Y = \sqrt{\frac{1}{n}\sum_{j=1}^{n}(y_j - \mu_Y)^2}, \qquad RMS_X = \sqrt{\frac{1}{n}\sum_{j=1}^{n} x_j^2} \tag{6}$$

$$D_k = \frac{1}{|Y|}\sum_{p}|Y_{k+1}(p) - Y_k(p)|, \qquad \text{centroid}(X) = \frac{\sum_f f\, M_X(f)}{\max(\sum_f M_X(f), 10^{-12})} \tag{7}$$

Here $M_X(f) = C^{-1}\sum_{c=1}^{C} |\,\mathrm{rFFT}(X_c)(f)|$ is the mean of the channel-wise magnitudes of the unwindowed, one-sided real FFT, including DC and the Nyquist bin when present. It is not a power or energy spectrum, nor the FFT of a channel-averaged waveform. Frequencies are in Hz at the canonical sample rate; the same magnitude weights define the spectral bandwidth. The denominator floor yields a zero centroid for silence. The illustrative audio atlas uses separately disclosed display-band aggregation; it does not replace this full-resolution metric computation.

Objective metric definitions I

The level count is the cardinality of the finite-value set after canonical quantization; an empty or non-finite artifact is invalid rather than assigned a default statistic. Every generated metric carries units, full-window and channel/transfer definitions, NumPy and implementation identity, and SHA-256 hashes of the computation/window definitions. The implementation fingerprint covers `metrics.py` and `artifacts.py`, not NumPy binaries, CPU behavior or a historical platform. These identify the calculation, not stimulus bytes (identified separately by canonical digest), calibration, or authorship. Observer interpretations remain separate typed claims.

Objective metric definitions I

For a future observer study, the estimand is defined by the protocol rather than by the generator itself:

$$\psi = \mathbb{E}[Y \mid \text{condition}, \ \text{protocol}] \tag{8}$$

This symbol names a planned analysis target only. DuckRabbit ships no human responses, fitted coefficients, or validated observer-level estimate.

Synthetic psychophysics boundary I

DuckRabbit's transparent synthetic-psychophysics diagnostic supports method orchestration. It is a hand-specified feature observer, not a pretrained vision model and not an empirical psychophysics substitute. Let $\phi(A)$ be the bounded feature vector extracted from a canonical artifact, w the code-owned feature weights, and $\tau > 0$ the declared temperature:

$$p_M(c \mid A_r, A_c) = \sigma\left(\operatorname{clip}\left(\frac{w^\top \phi(A_c) - w^\top \phi(A_r)}{\tau}, -60, 60\right)\right) \tag{9}$$

The enclosing diagnostic serializes model identity/version and fingerprint, actual feature values, weights, temperature, seed, calibration statement, canonical digests, numerical clipping/tie rules and `human_data=false`. Detached predictions retain scores and feature names but require that enclosing specification to reconstruct the model. Exact-self and all-zero-weight controls yield 0.5; ties choose the comparison deterministically.

Synthetic psychophysics boundary I

The default sweep also zeroes each declared feature weight and varies temperature by factors 0.5, 1 and 2. Five image features are active; nine other ablations and the 1-times temperature reproduce existing curves rather than independent evidence. The displayed min/max range covers only these declared configurations, including the null. It is model sensitivity, not an empirical confidence interval, a human psychometric function, or uncertainty over all possible models. Joint perturbations, preprocessing/quantization sensitivity, calibration and observer variation are not covered. Human work still requires task, consent, calibrated presentation, preregistration and actual response data.